Supplementary Material
Coordinated Multi-Robot Navigation with Formation Adaptation

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I. Implementation of Spring-Damper Models as Rewards

In our hierarchical learning method, we model both components using a reward functioned defined as follows:

$$R^{adapt} = \sum_{\mathbf{v}_i, \mathbf{v}_j \in \mathcal{V}} -\alpha |\mathbf{d}_{i,j} - \mathbf{p}_{i,j}| - (1-\alpha)\mathbf{q}_{i,j} \tag{1}$$

where α is a hyperparameter to balance the spring and damper components. This reward encoding the spring-damper model is used together with the classic rewards defined for reaching to the goal position and obstacle avoidance for training AFOR. For navigation, the reward function dynamically rewards robots based on their orientation and proximity to the goal, facilitating efficient navigation within the environment. For avoiding collisions, the reward function gives numerical penalty to discourage robots collide with an obstacle or another robot. The classic reward is computed as follows:

$$R_i^{classic} = \begin{cases} \mathbf{q}_i \cdot \frac{\vec{r}_{\text{goal}}}{\|\vec{r}_{\text{goal}}\|} \times \|\mathbf{q}_i\| & \text{if } \|\vec{r}_{\text{goal}}\| > 0.0, \\ 15.0 & \text{if } ||\mathbf{p}_i - \mathbf{g}_i||_2 < 0.1, \\ -1.5 & \text{if } ||\mathbf{p}_i - \mathbf{p}_j||_2 < 0.2 \end{cases} \tag{2}$$

where $\vec{r}_{\text{goal}}$ is the reward vector pointing towards the goal, representing the direction and magnitude of the reward based on the robot's current position relative to the goal, and the dot product $\mathbf{q}_i \cdot \frac{\vec{r}_{\text{goal}}}{\|\vec{r}_{\text{goal}}\|}$ computes the alignment of the robot's velocity vector with the direction to the goal, scaled by the velocity magnitude to reward faster movement towards the goal.

II. Experimental Setups

- In AFOR, the team of robots is represented as a graph, where the edge set is constructed using a radius graph based on the robots' communication range, which is set to 2 meters.
- For team coordination at the upper level, we use an encoder with weights $\mathbf{W}_z$ of dimension 6×64, followed by a GNN layer with weight matrix $\mathbf{W}_h$ of dimension 64×64. At the lower level, individual robot control is handled by a three-layer MLP with weight matrices of dimensions 64×64, 64×64, and 64×6, respectively.
- Our spring-damper model integrates a balance factor α, set to 0.6, to maintain equilibrium between the spring and damper forces.
- We implement Proximal Policy Optimization (PPO) to train the hierarchical model, with the PPO loss defined as the clipped objective to ensure stable updates. Formally, the PPO loss is defined as follows in the main paper:

$$\sum_{\mathbf{v}_i \in \mathcal{V}} \mathbb{E}_{\mathbf{s}_i, \mathbf{a}_i \sim d^{\pi_{\theta_{old}}}} \left[\min \left(\frac{\pi_\theta(\mathbf{a}_i|\mathbf{s}_i)}{\pi_{\theta_{old}}(\mathbf{a}_i|\mathbf{s}_i)} A^{\pi_{old}}(\mathbf{s}_i, \mathbf{a}_i), \quad \text{clip} \left(\frac{\pi_\theta(\mathbf{a}_i|\mathbf{s}_i)}{\pi_{\theta_{old}}(\mathbf{a}_i|\mathbf{s}_i)}, 1-\lambda, 1+\lambda \right) A^{\pi_{old}}(\mathbf{s}_i, \mathbf{a}_i) \right) \right] \tag{3}$$

  where the clipping range λ is set to 0.95, with a learning rate of 5×10^{-5} and a discount factor γ of 0.995.
- During experiments, we obtain the velocity $\mathbf{q}_i$ and position $\mathbf{p}_i$ of each robot using Gazebo for simulations and an OptiTrack motion capture system for real-world trials.
- Training is performed in a custom PyGame-based environment simulating multi-robot navigation, where randomized starting and goal positions are assigned to the robots. Both upper level and lower level policies are trained simultaneously over 700 episodes to ensure effective coordination and control.

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III. Examples of Computing the CFI Evaluation Metric

As mathematically defined in the main paper, Contextual Formation Integrity (CFI) is computed as follows:

$$\beta \left(1 - \delta^{-1} \min\left(\left|W - (\tau + \delta)\right|, \left|W - (\tau - \delta)\right|\right)\right) + (1 - \beta)\epsilon$$

The first term of CFI assesses the gap usage, where W denotes the team formation's maximum width, τ is a safety threshold computed as the corridor width minus a safety margin, and δ represents the rigidness of the formation with a smaller value indicating stricter formation requirements. The second term $\epsilon \in [0, 1]$ assesses the team's shape integrity. Putting the two terms together, CFI evaluates how well a robot team utilizes the corridor space and maintains formation integrity, balanced by the coefficient $\beta \in [0, 1]$. CFI $\in [0, 1]$, with higher values indicating better performance. In our experiments, we set τ to twice the robot's width and $\beta = 0.5$. For a team of n robots, the CFI is computed as follows:

- *Wedge formation*: $\epsilon = 1 - \frac{2|L_l - L_r|}{L_l + L_r} - \frac{|2L_m - L_b|}{L_b}$, where L_l, L_r, L_m, L_b represents the lengths of the left, right, middle, and base sides of the isosceles triangle formed by the robots, respectively.
- *Circle formation*: $\epsilon = 1 - \frac{1}{n}\sum_{i=1}^{n} \frac{\theta_i}{\frac{(n-2)\times 180}{n}}$, where θ_i represents the interior angle of the triangle formed with the i-th robot as a vertex, and $\frac{(n-2)\times 180}{n}$ corresponds to the interior angle of a polygon, which approximates a circle.
- *Line formation*: $\epsilon = 1 - \frac{1}{n-1}\sum_{i=1}^{n-1} \frac{(n-1)L_{i,i+1}}{L}$, where $L_{i,i+1}$ represents the distance between each neighboring robots, L denotes the full width of the robot team.

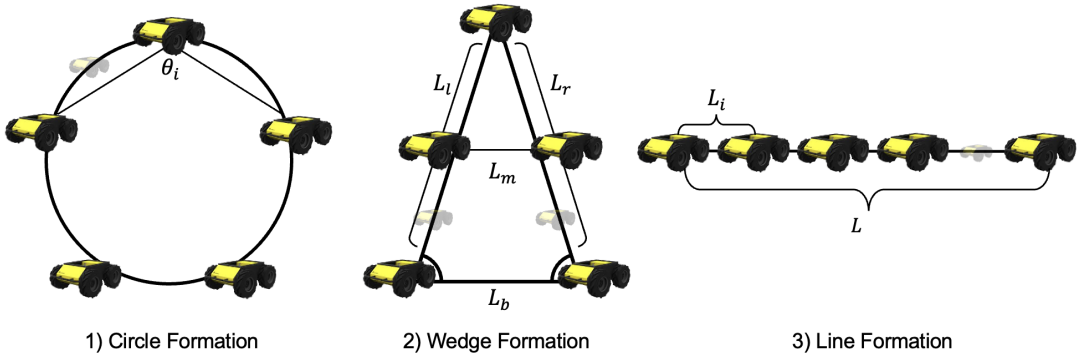


Fig. 1. Visual representation of how CFI metric evaluates three formations with different number of robots.

IV. Case Studies in High-fidelity Simulations across different scenarios

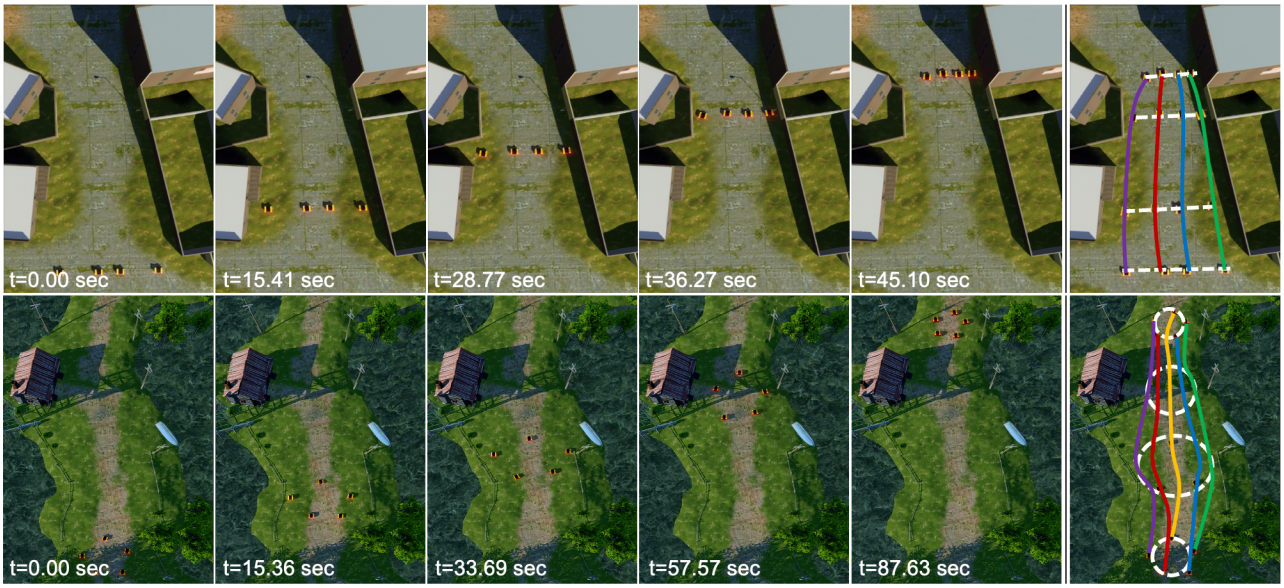


Fig. 2. Qualitative results on formation adaptation using a team of Warthog robots in a high-fidelity multi-robot simulation. The robots adaptively maintain a line and circle formation when navigating through a long and narrow corridor.

We conducted additional experiments using a high-fidelity simulator. In contrast to the Husky robots used in our main paper, we deployed different types of unmanned ground vehicles, specifically Warthog robots, in various scenarios for further testing. As shown in Figure 2, a team of four robots in a line formation navigates around buildings, while a team of five robots traverses an outdoor field, both scenarios involving narrow corridors. The qualitative results demonstrate that our approach enables teams of varying sizes to successfully navigate while maintaining adaptive formation shapes suited to the environment.